

**E-COMMERCE CUSTOMER RETENTION
& COHORT ANALYSIS**

Final Project Report

Methods, findings, recommendations and delivery evidence

This report summarizes the implemented portfolio project and its evidence. It documents customer purchasing behavior, measurement limitations and proposed business actions. No marketing intervention has been deployed and no realized uplift is claimed.

Evidence base	Edition
UCI Online Retail II; historical UK retail	22 September 2026
Final repository; metric contract 1.1	One combined, executed notebook
14-step implemented analytical scope	English portfolio documentation

Prepared for learning, technical review and portfolio presentation. The Vietnam marketplace context motivates the analytical questions; the observed results describe the historical source retailer.

Executive summary

The project examines how an e-commerce team can prioritize second-purchase, reactivation and customer-service initiatives using transaction history. It applies SQL and Python to UCI Online Retail II, a historical UK gift retailer with wholesale activity. The original Vietnam marketplace framing guides the questions; findings remain specific to the source data.

Measure	Validated result
Source / eligible lines	1,067,371 / 972,818
Eligible invoices / identified accounts	38,615 / 5,821
Complete reporting period	December 2009 through November 2011
Gross merchandise sales	GBP 18,933,017.23 before credits
Identified sales coverage	87.25%
90-day repeat purchasing	41.14%: 1,452 of 3,529 mature primary customers
Top-decile historical concentration	583 accounts contribute 63.56% of identified sales
Historical-repeat reactivation pool	1,728 accounts inactive for more than 90 days

The analysis supports a measured second-purchase trial, a separate reactivation trial, monitoring of large-account concentration and better customer-linkage measurement. First-basket differences generate hypotheses; geography does not support a clear budget preference after checking cohort mix.

The final repository includes code, seven SQL files, one executed notebook, eleven aggregate CSVs, six charts, an HTML dashboard and reports. The 14-step implemented scope is complete and the final packaging rebuild passed. Certain original proposal ideas were narrowed or omitted, as documented later in this report.

Repository evidence: `reports/case_study.md`; `reports/final_delivery_validation.json`

1. Business context and decision scope

The original proposal sought a job-ready customer analytics portfolio relevant to a fictional Shopee/Lazada-style marketplace and data analyst work in Ho Chi Minh City. Its central concern was whether growth relies heavily on new customers and promotions, and where retention resources should be focused. The concern was a scenario, not an observed fact about the source retailer.

The implemented business question is: how should an e-commerce team prioritize retention initiatives using repeat purchasing, cohort performance and observed customer value? Growth and CRM teams need a second-purchase and reactivation measurement framework; commercial teams need customer and first-basket comparisons; managers need consistent operating indicators.

Decision area	Analytical evidence
Acquisition versus returning activity	Monthly purchasing accounts, new/returning composition, invoices and gross sales
Second-purchase opportunity	30/60/90-day repeat rates, complete follow-up and conditional purchase timing
Customer-age behavior	Calendar-month cohort tables, maturity-aware pooling and heatmap
Value and service exposure	RFM, historical concentration and fixed-window observed spending
Targeting hypotheses	First-basket and country comparisons, group sizes and cohort-mix checks
Action evaluation	Randomized trial proposals, defined outcomes, owners and guardrails

The work demonstrates a complete analytical process rather than an operational marketing system. Customer IDs are historical accounts, contemporary contactability is unknown, and no campaign deployment, contact list or future revenue forecast is delivered. Recommendations require current business data before execution.

Report structure: data and quality; measurement and model; monthly and retention evidence; RFM/value and group comparisons; recommendations; reproducibility; proposal alignment and references.

Repository evidence: [reports/project_plan.md](#); [reports/dataset_selection.md](#)

2. Data source and analytical model

UCI Online Retail II contains 1,067,371 item lines in two worksheets, covering December 2009 through December 9, 2011. The original workbook is preserved and verified by SHA-256. The source records invoice reference, stock code, description, quantity, invoice date, unit price, customer identifier and country.

The reporting cutoff is December 1, 2011, exclusive, giving 24 complete months. Full earlier history is retained, but headline first-observed cohort comparisons begin March 2010 after an initial history period. A first observed purchase is not necessarily a newly acquired customer in the real business.

Analytical table	Grain / rows
stg_order_lines	One source row with raw values, flags and disposition; 1,067,371
fact_order_lines	One eligible merchandise line; 972,818
fact_orders	One eligible invoice; 38,615
customer_features	One identified historical account; 5,821
customer_month_activity	One purchasing customer-month; 24,850

Invoice and customer histories support downstream customer_purchase_history, customer_repeat_windows, monthly_kpis, cohort_retention, RFM, customer_value and comparison populations. Product-role mapping is a separate lookup. Sensitivity views preserve an alternate duplicate treatment without mixing it into primary data.

The model implements explicit analytical grains and relationships in DuckDB. It does not reproduce every dimension table in the proposal's recommended star schema. The public folder contains aggregate results; detailed database and Parquet files are generated locally.

The source lacks native acquisition channel, product category, discount, payment, delivery, margin and Vietnam district fields. No synthetic replacement fields were introduced. GBP remains the financial unit.

Repository evidence: data/raw/source_manifest.json; sql/01b_clean_data.sql; data_dictionary.md

3. Data quality and cleaning decisions

Cleaning applies a documented contract, retaining raw lineage and all applicable exclusion reasons. An additive primary disposition prevents overlapping quality flags from being double-counted. There are 972,818 included and 94,553 excluded rows.

Issue	Implemented treatment and evidence
Worksheet overlap	Compare full normalized invoice multisets; retain later identical version. Exclude 22,523 older-copy rows. Conflicts are quarantined.
Repeated retained lines	Keep earliest identical row in primary analysis; exclude 11,812 rows; retain duplicate-inclusive sensitivity views.
Header conflict	Quarantine inconsistent timestamps/customer headers; 5,721 rows fall into this primary disposition after earlier exclusions.
Price precision	Bounded six-decimal normalization only within 1e-9 GBP; preserve raw values and flags.
Product roles	5,131 codes mapped: 4,720 merchandise, 25 non-merchandise, 386 unresolved. Only classified merchandise enters primary purchases.
Credit and invalid records	Require positive purchase quantities/prices and valid invoice/date. Credit records remain auditable; no unverified purchase-refund matching.
Missing identity	Keep eligible anonymous invoices in sales; exclude them from customer metrics.

A total of 243,007 raw lines lacked customer IDs. After cleaning, identified orders represent 92.58% of invoices and 87.25% of gross sales. Missing identity can change who is represented in retention analysis.

Unresolved product exposure includes 846 otherwise eligible lines worth GBP 338,725.57, about 1.79% of retained sales. It remains excluded and disclosed because the meaning of manual/sample/image entries is not established. It is not an estimate of lost business.

Repository evidence: [reports/data_quality_report.md](#); [reports/cleaning_report.md](#); [src/product_roles.csv](#)

4. Measurement contract

Measure	Definition and interpretation
Gross merchandise sales	Sum of eligible positive merchandise purchases, before credit adjustments. Includes unidentified invoices; does not establish profit or net revenue.
Monthly active / new / returning	Distinct purchasing accounts. New means first observed purchase month; returning means first observed earlier. New + returning = active.
D-day repeat purchasing	Eligible if first purchase + D days < cutoff. A second distinct invoice at or before that endpoint counts. Incomplete windows excluded from both counts.
Monthly cohort retention	Distinct cohort accounts purchasing in month k / original cohort size. Month 0 = 100%; future months null; observed no-purchase months zero.
RFM	Snapshot 2011-12-01; recency from full-history last purchase; F/M from trailing 12 months; tie-preserving scores; dormant retained.
Historical concentration	Top ceil(10% of identified accounts) by historical gross sales / all identified historical gross sales.
Reactivation pool	At least two historical purchases and recency strictly greater than 60, 90 or 120 days. Operational candidates, not confirmed churn.

The common 90-day population makes 30/60/90-day repeat progression comparable. Own-window rates use different mature populations. Calendar-month retention and elapsed-day repeat rates answer different questions and cannot be substituted.

Group comparisons freeze first-order attributes, require equal 90-day follow-up and suppress headline rates below $n=100$. Raw intervals use 95% Wilson limits. Cohort standardization applies shared month weights and minimum sample rules; it addresses acquisition timing only.

Repository evidence: `data_dictionary.md`; `src/metric_spec.json`; `reports/metric_examples.md`

5. Monthly operating baseline

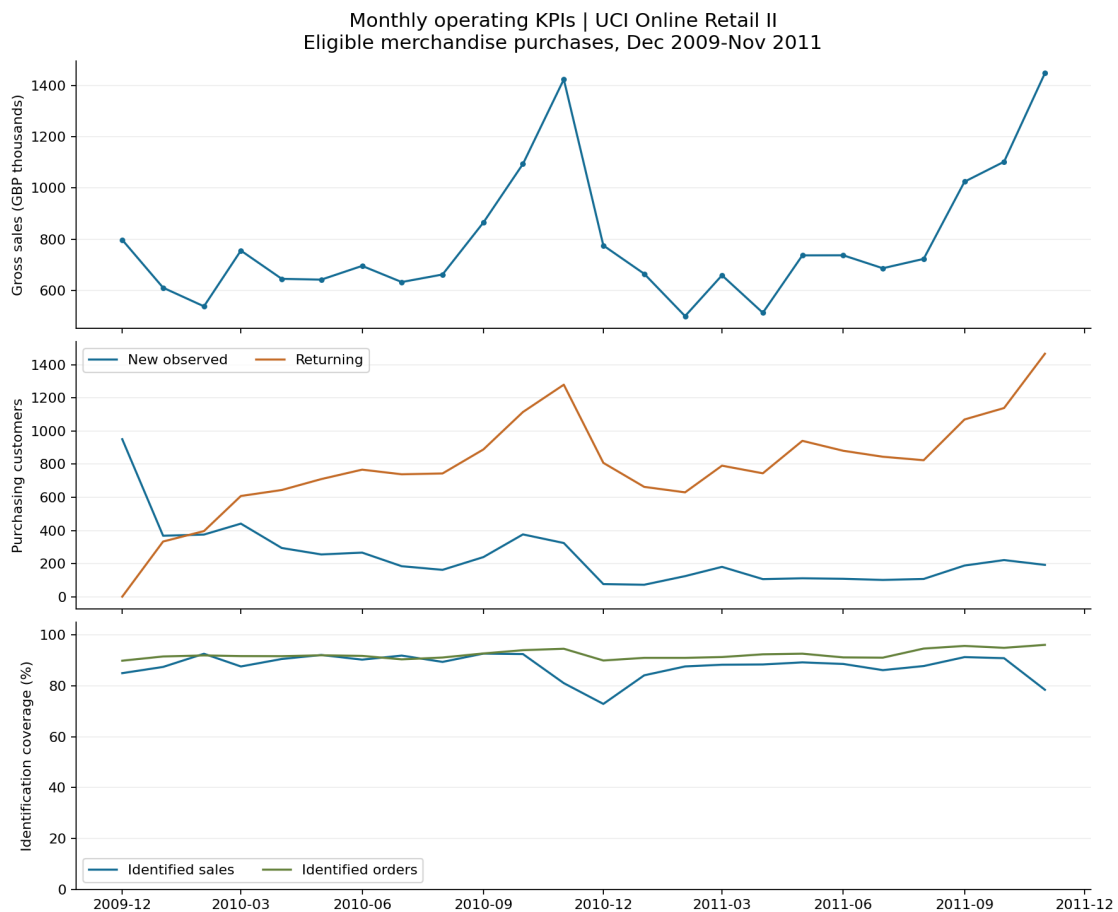


Figure 1. Monthly operating indicators under the complete-month reporting scope. Source: monthly_kpis.csv and the validated invoice population.

Across 24 complete months, eligible invoices generate GBP 18,933,017.23 gross merchandise sales with overall AOV of approximately GBP 490.30. Identified sales total GBP 16,518,355.99; the remaining GBP 2,414,661.24 belongs to 2,864 unidentified invoices.

Monthly new/returning composition provides operating context. A buyer with multiple invoices in their first observed month remains a new customer for that month. Customer counts describe purchasing accounts, not website or app engagement.

The calendar spine preserves empty observed months, and undefined ratios remain missing. A sum of monthly active customers double-counts accounts active in several months; unique historical customers must be counted from customer identities or first-observed arrivals.

Repository evidence: data/processed/monthly_kpis.csv; reports/monthly_kpis_report.md

6. Repeat purchasing and timing

Follow-up window	Repeaters	Eligible accounts	Rate
30 days	690	3,529	19.55%
60 days	1,158	3,529	32.81%
90 days	1,452	3,529	41.14%

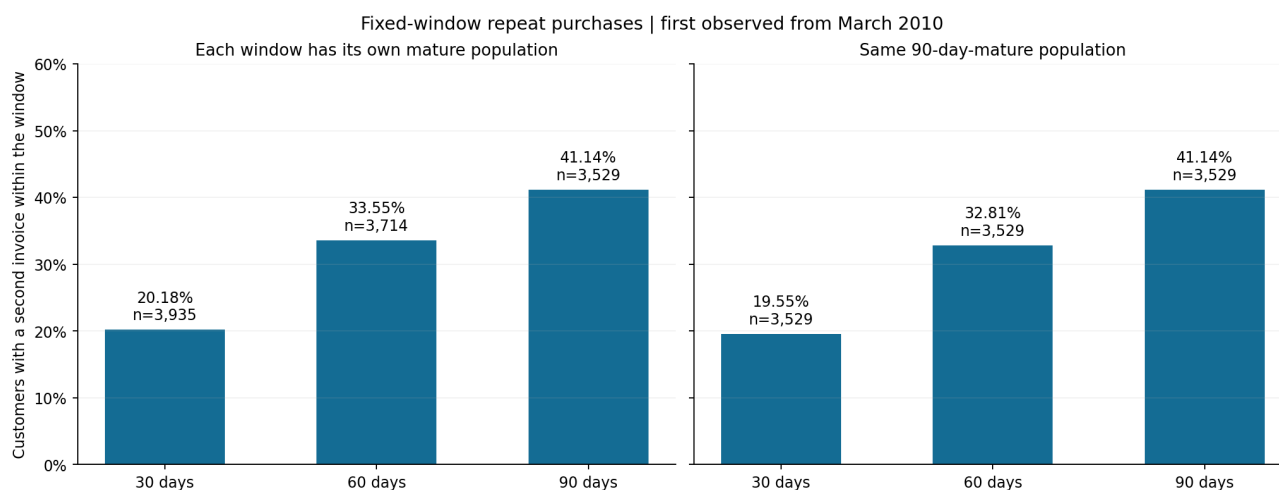


Figure 2. Fixed-window repeat purchasing. The common-90-day panel uses the same mature customers across all windows.

Among the 1,452 who repeated within 90 days, median time to the second invoice was 32.00 days. This conditional value excludes nonrepeaters; it does not establish the optimal CRM contact date. Same-timestamp distinct invoices are retained in the primary definition, with a collapsed-occasion sensitivity check.

The measured repeat rate creates a clear experimental outcome. A modern second-purchase program should compare randomized groups after full follow-up rather than treating this historical rate as an achieved improvement.

Repository evidence: `data/processed/repeat_purchase_summary.csv`; `reports/retention_analysis.md`

7. Calendar-month cohort retention

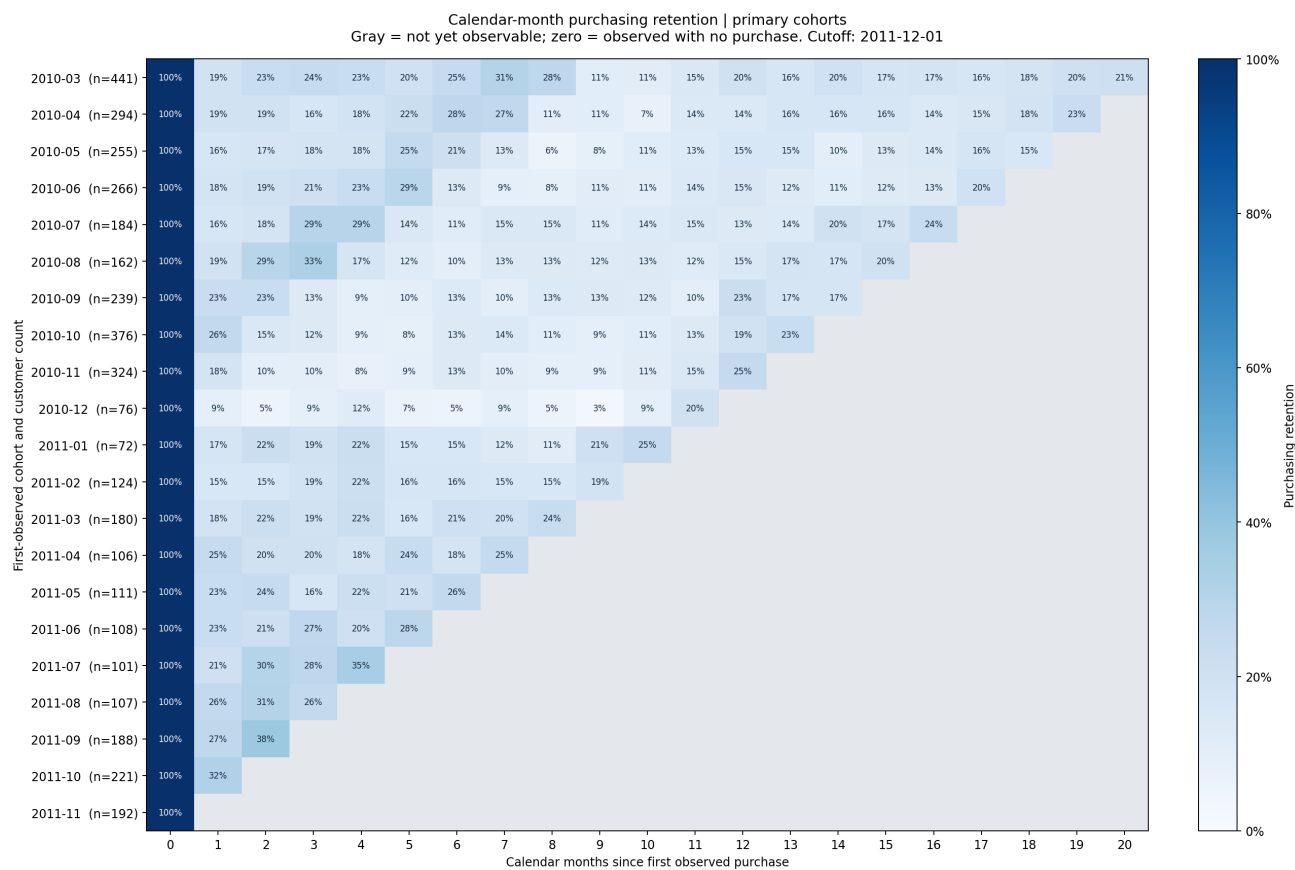


Figure 3. Primary first-observed cohorts from March 2010. Gray cells are unobserved future months; cohort labels show population size.

The heatmap tracks purchasing in each calendar month after the first observed purchase month. A customer contributes once per cohort-month even if they place several invoices. A customer may skip a month and return, so the series is not required to decline monotonically.

The full grid contains 576 cohort-month cells, of which 276 are future/unobserved. The primary visualization excludes initial-history cohort rows while retaining that history to identify first purchases correctly. Pooled Month-k rates use only mature in-scope cohorts and weight by cohort size.

The figure supports monitoring and hypothesis formation. Differences can reflect seasonality, acquisition timing, group size, wholesale composition or coverage. The chart alone does not identify why customers return or establish an intervention effect.

Repository evidence: `data/processed/cohort_retention.csv`; `data/processed/cohort_retention_matrix.csv`; `data/processed/cohort_retention_pooled.csv`

8. RFM segmentation

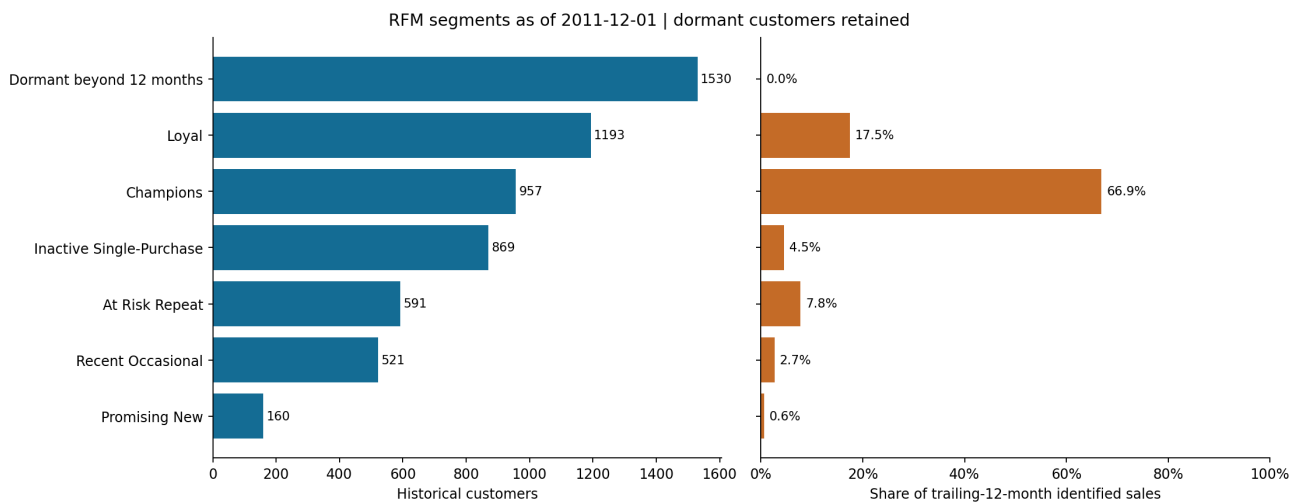


Figure 4. Rule-based customer segments at the December 1, 2011 snapshot. Frequency and money use the trailing year.

Segment	Accounts	Trailing-year sales share
Dormant beyond 12 months	1,530	0.00%
Loyal	1,193	17.48%
Champions	957	66.90%
Inactive Single-Purchase	869	4.55%
At Risk Repeat	591	7.78%
Recent Occasional	521	2.66%
Promising New	160	0.63%

All 5,821 historical identified accounts remain represented. Dormant accounts have zero trailing-year F/M and null scores. Scores preserve ties; segments follow first-match rules. Champions' high spending share partly follows from their definition. Labels are operational descriptions, not verified psychological states or predictions of permanent churn.

Repository evidence: `data/processed/rfm_segments.csv`; `sql/05_rfm_segments.sql`

9. Customer value and sensitivity

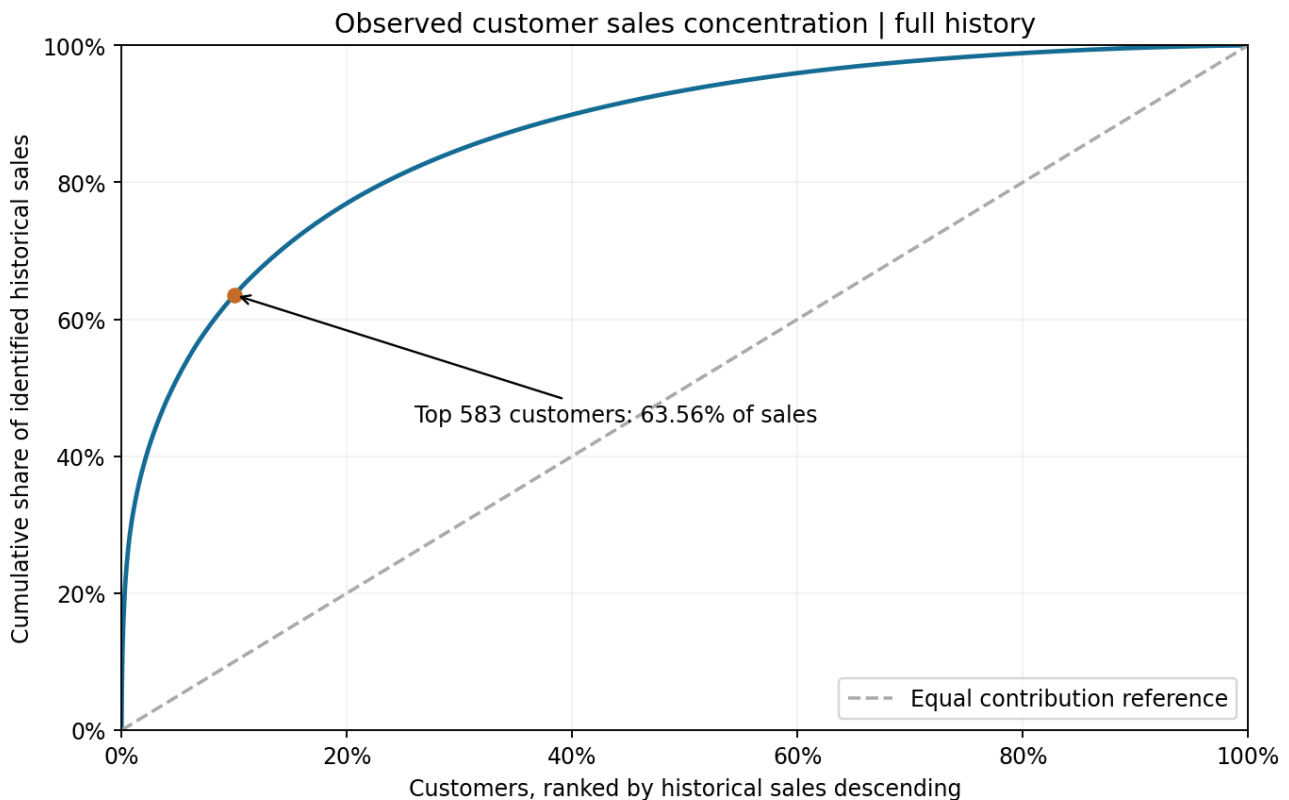


Figure 5. Cumulative share of identified historical sales, with customers ranked by their observed gross spending.

The top 583 accounts contributed GBP 10,499,700.03, or 63.56% of identified historical sales. This reveals exposure to a relatively small group of large accounts. It does not establish contribution margin, future spend or which accounts respond best to an incentive.

For 3,529 mature primary customers, first-90-day observed sales have mean GBP 682.83 and median GBP 369.02, including the initial purchase. Equal follow-up improves comparability, but these are observed-window values rather than predicted lifetime value.

Keeping within-sheet duplicate lines changes top-decile share to 63.48%; excluding the extreme invoice 541431 changes it to 63.42%. The alternatives change four and one RFM segment assignments, respectively. The concentration pattern persists under these two checks, while unknown refunds, margin and other quality limitations remain.

Repository evidence: `data/processed/rfm_sensitivity.csv`; `reports/customer_value_and_segments.md`

10. Reactivation and first-order differences

Inactivity threshold	Historical-repeat candidates	Their historical gross sales
More than 60 days	2,071	GBP 3,754,788.85
More than 90 days	1,728	GBP 2,917,464.33
More than 120 days	1,543	GBP 2,455,693.26

The primary reactivation rule uses historical repeat frequency and includes dormant repeaters. It therefore produces 1,728 candidates, compared with 591 in the trailing-year At Risk Repeat RFM segment. The associated money is past sales, not forecast recoverable revenue.

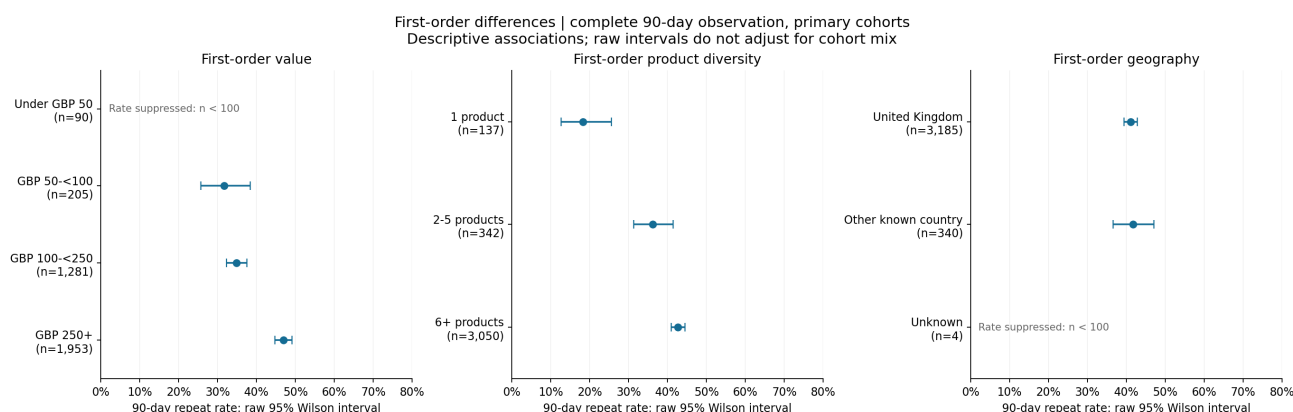


Figure 6. First-order groups on equal 90-day follow-up. Intervals are raw 95% Wilson intervals; small groups are suppressed.

GBP 250+ first baskets repeat at 46.95% (917/1,953), versus 34.89% (447/1,281) for GBP 100-<250. Six-plus-product baskets repeat at 42.72%, versus 18.25% for one product. Basket size, intent and wholesale behavior may overlap. These associations suggest experiments, not a causal claim that increasing basket size improves retention.

Repository evidence: data/processed/reactivation_thresholds.csv; data/processed/customer_comparisons.csv

11. What group comparisons cannot establish

All group comparisons use the mature primary population and first-invoice features. Raw Wilson intervals quantify binomial uncertainty within each reported group; they are not simultaneous multiple-comparison intervals or causal uncertainty estimates.

Comparison	Cohort-mix assessment
First-basket value	Three shared qualifying months, but the restricted GBP 50-<100 group has only 64 eligible customers. Adjusted rates withheld for the feature.
Product diversity	No shared month satisfies the minimum sample for all compared groups. No adjusted product ranking published.
Country	Eight shared months satisfy the rules. UK/other-known raw rates 41.10%/41.76% become 41.77%/39.82% after common weighting.

The geography ordering changes after a limited adjustment for cohort timing. The small raw difference and overlapping intervals do not justify selecting countries for acquisition or retention budget on this evidence. Standardization does not adjust every difference in customer intent or commercial mix.

A joint basket/product breakdown retains an association within six-plus-product baskets: 35.33% repeat for GBP 100-<250 versus 47.39% for GBP 250+. This is a descriptive conditional comparison with remaining confounding. Small joint cells are suppressed.

Publishing withheld comparisons is part of the result: the data do not support every desired adjusted ranking. Additional contemporary data on channels, categories, discounts, service experience and margins would be needed to investigate the original broader hypotheses.

Repository evidence: data/processed/comparison_by_cohort.csv; data/processed/basket_value_product_comparison.csv; reports/customer_analysis_summary.json

12. Recommendations and evaluation plan

Priority / owner	Action and evaluation
0. Analytics and data operations	Investigate missing IDs and unresolved product roles. Monitor identified-sales/invoice coverage, unresolved exposure and sampled identity accuracy. Record population changes.
1. CRM with analytics	Randomize eligible contemporary first purchasers into follow-up and business-as-usual holdout. Stratify by first-basket value and enrollment period. Primary outcome: 90-day intention-to-treat repeat-rate difference with confidence interval.
2. CRM	Refresh the historical-repeat and >90-day inactivity rule. Run a separate randomized reactivation trial within recency/value strata. Primary outcome: purchasing within 60 days of assignment.
3. Commercial and customer operations	Monitor large-account concentration and service issues. Test appropriate service support against standard service; measure sales per assigned account and purchasing in a predefined window.

For the second-purchase pilot, secondary outcomes include 90-day gross sales per assigned customer and purchase timing. Guardrails include unsubscribes, returns, contact cost and contribution margin where available. Set sample size from the actual eligible baseline, minimum worthwhile effect, significance and power before launch.

The reactivation clock begins at assignment, unlike the first-purchase retention clock. Use a holdout to separate natural return from campaign response, and avoid overlapping assignments across trials. Historical account counts illustrate the method; they are not a usable modern contact list.

Allow enrollment plus complete outcome follow-up. Do not stop because an interim result looks favorable. A treatment-minus-control estimate, costs and guardrails are required before describing an intervention as successful. No campaign uplift or ROI has yet been measured.

Repository evidence: [reports/case_study.md](#)

13. Reproducibility and verification

The final package was rebuilt from the verified original workbook in a separate output directory. All nine pipeline scripts and the combined notebook completed successfully. The notebook has 17 executed code cells with no error outputs. Six synthetic regression suites passed.

Check	Recorded evidence
Analytical regressions	Parsing, overlap multiplicity, cleaning/identity rules, date/window boundaries, cohort future cells, RFM ties, suppression and failure handling
Final result comparisons	All eleven published CSVs and all six PNG charts byte-identical to the prior validated results
Earlier full database comparison	18 tables and two views reconciled by schema and bidirectional row-multiset comparison
Presentation and packaging	155 local links checked; public Git exceptions verified; one notebook and aggregate results included
Earlier environment/source audit	New Python 3.12.14 environment installed from cached distributions; original workbook freshly downloaded and hash-verified
Known runtime note	Sandbox IPython history storage emitted a write warning; analysis execution completed and notebook outputs had no errors

The final recorded run uses Windows and pinned direct dependencies. Transitive packages remain unlocked; other operating systems and every future dependency combination were not tested. Network source availability can change. The record supports the reviewed workflow and data, not a claim of universal bug absence.

The reports are curated narratives. A source or definition change requires editorial review after regeneration. The repository has not been automatically published to GitHub. The dashboard is local HTML, not a Power BI file or hosted deployment.

Repository evidence: reports/pipeline_run.json; reports/reproducibility.md; reports/final_delivery_validation.json

14. Proposal alignment and delivered scope

Original proposal area	Final disposition
SQL/Python, cleaning and reproducibility	Implemented with explicit grains, a metric contract, independent validation and documented execution.
Monthly KPIs, repeat and cohorts	Implemented. Repeat windows are 30/60/90 days. Optional 180-day expansion was not implemented.
RFM, value, inactivity	Implemented using historical/fixed-window observed value and operational rules; no predictive CLV or churn model.
Category/discount/service/local analysis	Unavailable fields prevented these analyses. First-basket and country comparisons were delivered instead.
Recommended star schema	Implemented invoice/line/customer/activity tables, rather than all suggested dimension tables.
Time-to-second-order visualization	Conditional timing analysis and median delivered; no separate distribution chart.
Power BI and dashboard	Power BI was optional in the proposal. A portable aggregate HTML dashboard was delivered.
Notebook and repository presentation	Three sections combined into one notebook at the user's request; public CSVs consolidated under data/processed.
Case study, recommendations, README	Delivered with quantitative findings, evaluation plans, limitations and reproduction instructions.

Completion refers to the later 14-step implementation scope. The initial proposal was broader, and its changes are made explicit here. The project remains aligned with the original purpose: using transactional data to measure customer behavior and support practical decisions.

The source's age, geography, gift/wholesale mix, account identifiers, missing identity, unresolved records and unverified refund linkage constrain interpretation. Current Vietnamese marketplace conclusions require new data and validation. These constraints belong beside findings in any portfolio presentation.

Repository evidence: `reports/project_plan.md`; `README.md`; `data_dictionary.md`

15. Deliverables and source references

Deliverable	Reviewer entry point
Project overview and execution	README.md
Definitions and teaching cases	data_dictionary.md; reports/metric_examples.md
Data preparation and analysis code	src/; sql/; src/checks/
Single analytical walkthrough	notebooks/ecommerce_retention_analysis.ipynb
Published tables	Eleven CSVs and README in data/processed/
Figures and interactive summary	images/; dashboard/index.html
Business narrative	reports/case_study.md
Technical assurance	reports/reproducibility.md; pipeline_run.json; final_delivery_validation.json

Primary data citation: Chen, D. (2012). Online Retail II [Dataset]. UCI Machine Learning Repository. DOI: 10.24432/C5CG6D. Dataset page: <https://archive.ics.uci.edu/dataset/502/online+retail+ii>. The repository attributes the dataset under CC BY 4.0. The original workbook is unchanged; transformations and exclusions are documented.

Planning source: E-Commerce Customer Retention & Cohort Analysis - Project Proposal, five pages supplied with the project. Its Vietnam marketplace context, proposed metrics, recommended structure and success criteria were compared with the implemented scope.

Implementation evidence: final repository edition dated September 22, 2026, metric contract 1.1. This report reproduces validated figures from the CSV/JSON outputs and uses the six project charts. Percentages and display money are rounded; full-precision values remain in the source artifacts.

Portfolio presentation: describe the project as an independent historical retail analysis demonstrating retention methods. State the dataset size, analytical techniques and quantified observations. Attribute assistance honestly and be prepared to explain the definitions and code. Do not describe proposed actions as implemented commercial gains.

These two companion PDFs are documentation deliverables. They do not change the analytical repository or its previously recorded 73-file validation inventory.